

AI Critique

The AI was useful for quickly converting the assignment into an executable JMeter design, but its first outputs were incomplete in three important ways. First, it treated FR-01, FR-07, and FR-17 as a natural single-user journey, even though coupon management is an administrator operation and the selected features do not provide a suitable read-heavy flow. Human review corrected this by separating coupon setup and teardown from the customer workflow and adding product reads. Second, the generated JMeter plan placed a random timer at Thread Group scope. JMeter therefore applied the delay before every sampler instead of once per business iteration, reducing a ten-second smoke run from 142 to only 26 samples. Third, the initial CSV contained 250 identities while the Spike test used 600 virtual users. Recycled identities caused duplicate registrations, and failed authentication produced thousands of misleading 403 responses. The corrected plan uses 1,000 rows and prevents the business flow from running without a successfully extracted token. The AI also overinterpreted short-run memory growth as proof of a leak and suggested a conventional database connection pool even though the SUT uses embedded SQLite. These failures occurred because the model reasoned from generic performance-testing patterns without executing JMeter scope semantics, checking data cardinality against peak concurrency, or validating the actual database architecture. The main lesson is that AI output should be treated as a testable hypothesis. Collaboration is reliable only when every design decision is exercised with a small run, raw logs are preserved, derived metrics are independently recalculated, and recommendations are checked against the implementation before acceptance.